

Miroslav Šotek

Independent researcher · Founder · Systems engineer

Switzerland · anulum.li · [GitHub](https://github.com) · [ORCID 0009-0009-3560-0851](https://orcid.org/0009-0009-3560-0851) · [PyPI](https://pypi.org/user/protoscience) · protoscience@anulum.li

Profile

Founder and engineering lead working across evidence-governed AI, multi-agent infrastructure, scientific computing, neuromorphic systems, quantum simulation, plasma physics and control. Builds systems from mathematical models through reproducible software, native acceleration, formal checks and hardware-facing paths.

Technical Scope

Languages: Python, Rust, TypeScript, JavaScript, Go, Julia, Mojo, C/C++, Lean 4, Verilog, SystemVerilog, WGSL, Shell and LaTeX.

Systems: LLM assurance, agent coordination, auditable memory, NLI/RAG, numerical solvers, PyO3/Maturin, SIMD, WebAssembly, WebGPU, FPGA RTL, formal verification, provenance and evidence packaging.

Quality: deterministic tests, coverage gates, strict typing where declared, Ruff, Cargo fmt/Clippy, CodeQL, Bandit, SPDX/REUSE, SBOMs, hash-pinned dependencies, preregistration and replayable result packs.

Selected Work

- **Synapse Channel:** local-first coordination bus for agent fleets with claims, durable messaging, audit and federation.
- **Director-AI:** response-level LLM factual-consistency and contradiction guardrail with NLI/RAG evidence.
- **Rigor Foundry:** evidence-bound repository inventory, audit-candidate generation and remediation planning.
- **SC-NeuroCore:** stochastic and neuromorphic stack spanning Python, Rust SIMD, HDC/VSA and RTL-generation workflows.
- **SCPN Fusion Core:** plasma physics, solver, validation and control research.
- **SCPN Quantum Control:** preregistered coupled-oscillator quantum experiments with hardware result packs.

Portfolio And Delivery

Maintains a mapped ecosystem of **39 portfolio repositories** across Reactor, Systems Integration and Control, Agentic Assurance, Neuromorphic Computing and Quantum Computing, plus the standalone HushLine utility. The public portfolio contains **34 project entry points** and the PyPI profile contains **19 published projects**, verified 2026-09-03.

Research Output

- At least 29 ORCID-linked Zenodo records as of 2026-09-03, including technical notes, preprints and software archives. See the [publication index](#).
- SCPN Framework v1.0 technical specification: [10.5281/zenodo.18507639](https://zenodo.org/record/18507639).
- Phase-system matched-false-alarm and grid regime-map preprints: [10.5281/zenodo.22113062](https://zenodo.org/record/22113062) and [10.5281/zenodo.22113116](https://zenodo.org/record/22113116).

Collaboration

Open to selective research, engineering, review, deployment and institutional partnership conversations. Commercial licensing and managed product work are available through ANULUM. Public GitHub currently marks the account as hireable.

Evidence boundary: registry publication does not imply peer review, acceptance, independent validation or production readiness. Last reviewed 2026-09-03.